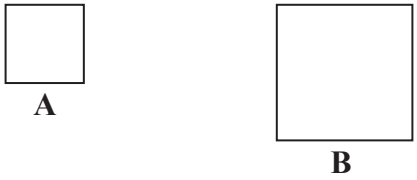
14 Here are two squares, **A** and **B**.



The length of each side of square **B** is 4 cm greater than the length of each side of square **A**.
The area of square **B** is 70 cm^2 greater than the area of square **A**.

Find the area of square **B**.
Give your answer correct to 3 significant figures.
You must show all your working.

Let square A have side x cm; B has side (x + 4) cm.

$(x + 4)^2 - x^2 = 70$; $8x + 16 = 70$.

$8x = 54$, so $x = 6.75$ and B side = 10.75 cm.

Area B = $10.75^2 = 115.5625 \approx 116$ cm^2 (3 s.f.).

4 marks total
Set area difference
Reduce to 8x=54
x=6.75
B area 116 cm^2

..... cm^2

(Total for Question 14 is 4 marks)

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